Supplementary materials

Effects of biological pretreatments of microalgae on hydrolysis, biomethane potential and microbial community

Mekdimu Mezmir Damtie[1,§], Jingyeong Shin[1,§], Hyun Min Jang[2], Hyun Uk Cho[3], Jinhua Wang[4], Young Mo Kim[1,*]

[1]*Department of Civil and Environmental Engineering, Hanyang University, Seongdong-gu, Seoul 04763, Republic of Korea*

[2]*Department of Environment & Energy and Soil Environment Research Center, Jeonbuk National University, Jeonju, Jeollabukdo 57896, Republic of Korea*

[3]*Department of Marine Environmental Engineering, Gyeongsang National University, Tongyeong, Gyeongnam 53064, Republic of Korea*

[4]*College of Resources and Environment, Shandong Agricultural University, Tai'an, 271018, China*

[§] *These authors equally contributed to this work.*

***Corresponding author.** Tel.: +82-2-2220-0314; fax: +82-2-2220-0399.
E-mail address: youngmo@hanyang.ac.kr

Table and Figure captions

List of table

Table S1 Model parameters for simulated methane yield.

List of figures

Figure S1 Effects of different types of pretreatment methods on environmental factors.

Figure S2 Measured and simulated methane yields pretreated microalgae.

Table S1 Model parameters for simulated methane yield.

Pretreatment type	HRT	Model Parameters			
		G_o	λ	R_{max}	R^2
Control (without pretreatment)	3	183.51	0.34	16.45	0.97
	2	128.76	1.32	9.97	0.99
	1	90.06	2.95	3.96	0.91
Anaerobic pretreatment at 85 °C	3	239.93	0.60	18.94	0.97
	2	166.21	0.15	17.35	0.96
	1	74.48	0.07	11.40	0.96
Anaerobic pretreatment at 55 °C	3	205.15	0.27	30.71	0.97
	2	189.11	0.18	18.85	0.97
	1	102.56	0.15	13.59	0.93
Aerobic pretreatment at 55 °C	3	194.24	0.06	29.17	0.98
	2	174.86	0.17	14.33	0.97
	1	106.17	1.84	5.49	0.92

Remarks:

- *Units and descriptions of the model parameters were lag-phase (λ, days),maximum methane yield (Go, mL/g-VS_{added}), and methane production rate (R_{max}, mL/g-VS_{added} /day).*
- *Initial methane yield data were duplicated. The error band for model parameters lies within 6%.*

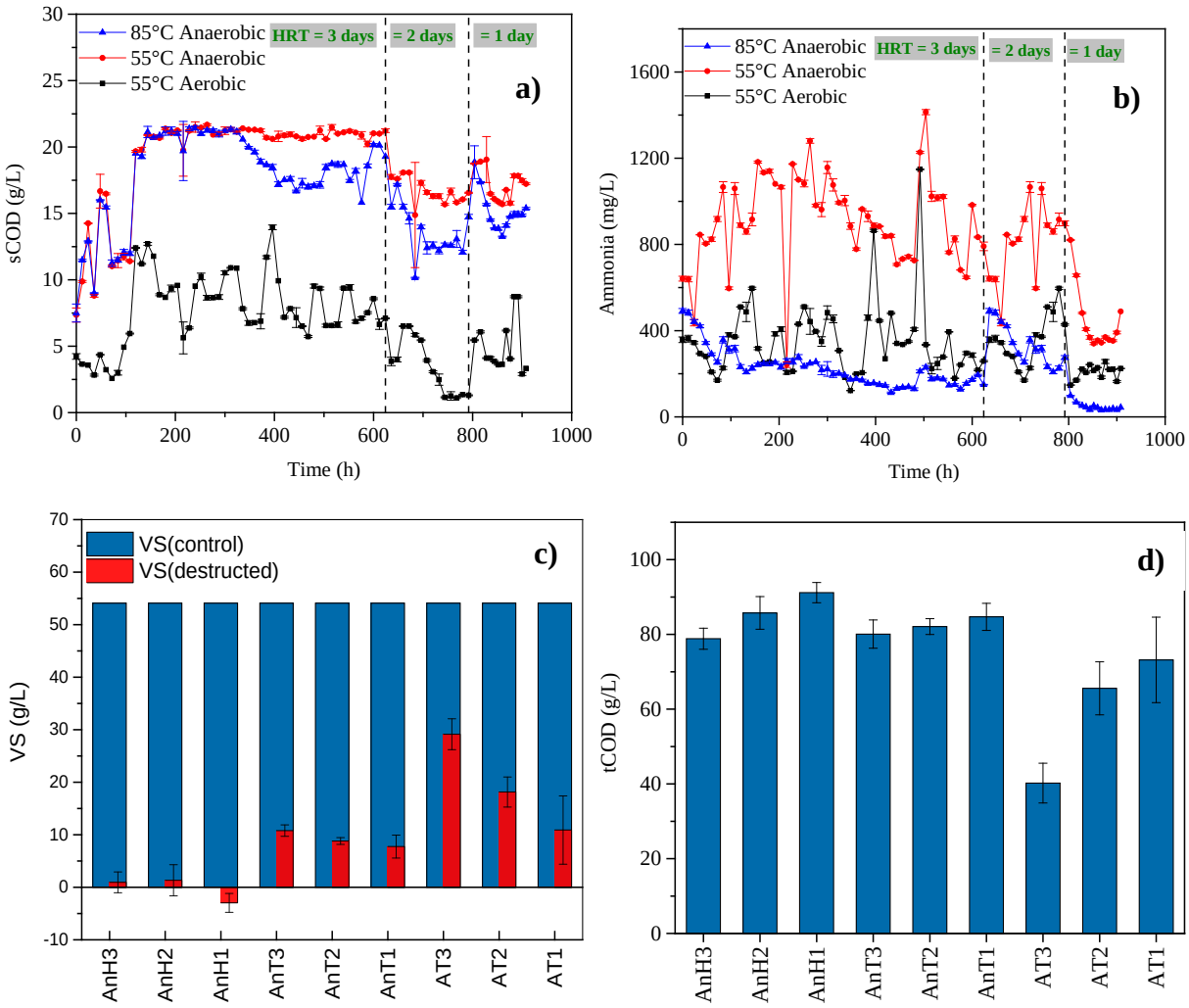


Figure S1 Effects of different types of pretreatment methods on (a) soluble COD; (b) release of ammonia gas; (c) volatile solids reduction; (d) total COD.

Note: *In the abbreviation AT1, AT2, AT3, AnT1, AnT2, AnT3, AnH1, AnH2, AnH3, "H" refers to hyperthermophilic temperature (85 °C); "T" represents thermophilic temperature (55 °C), "An" represents anaerobic pretreatment process, and "A" represent aerobic pretreatment process. The numbers 1, 2 and 3 refer to the HRTs(days).*

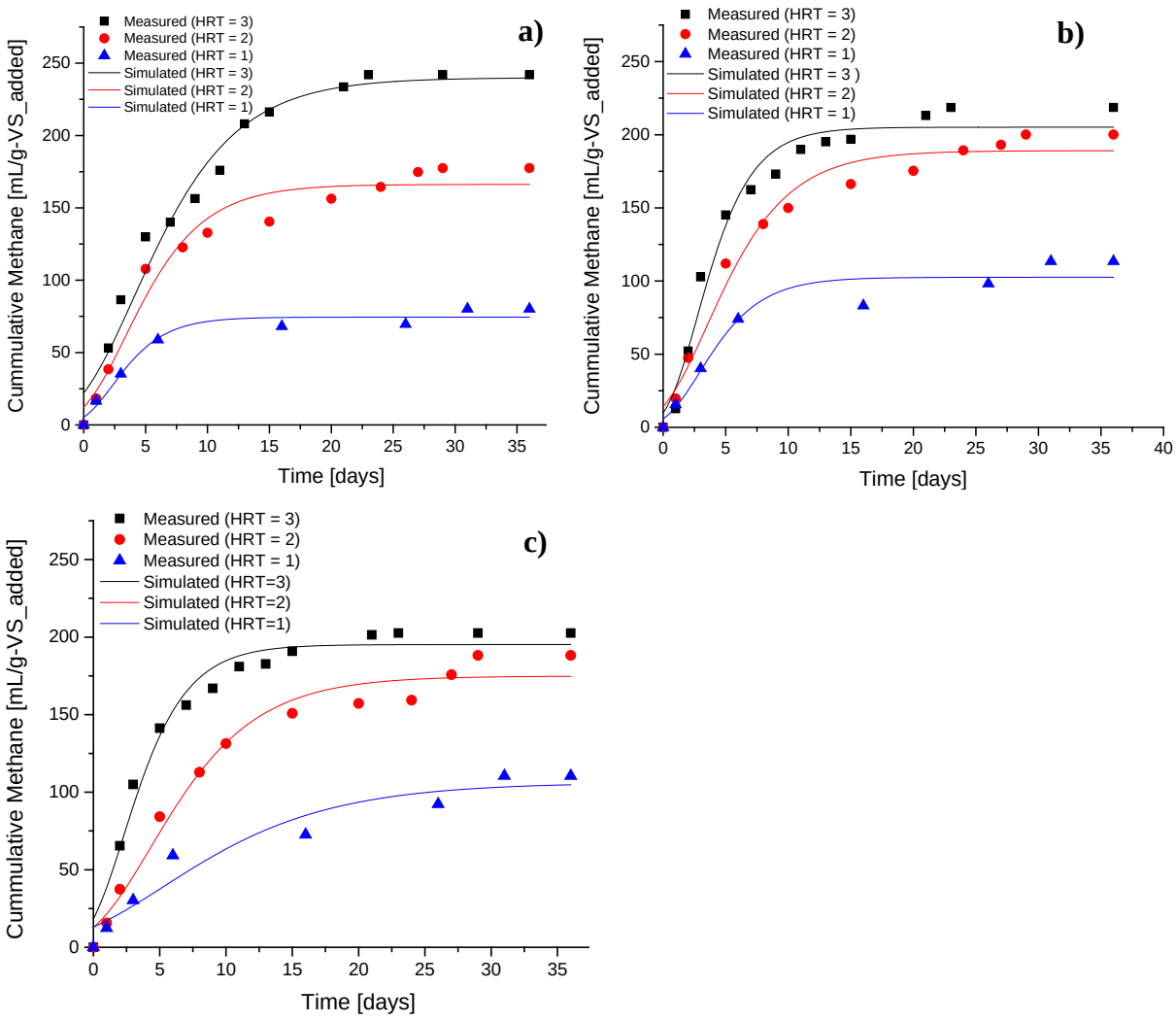


Figure S2 Measured and simulated methane yield of pretreated microalgae under (a) anaerobic pretreatment at 85 °C; (b) anaerobic pretreatment at 55 °C; (c) aerobic preatment at 55 °C.